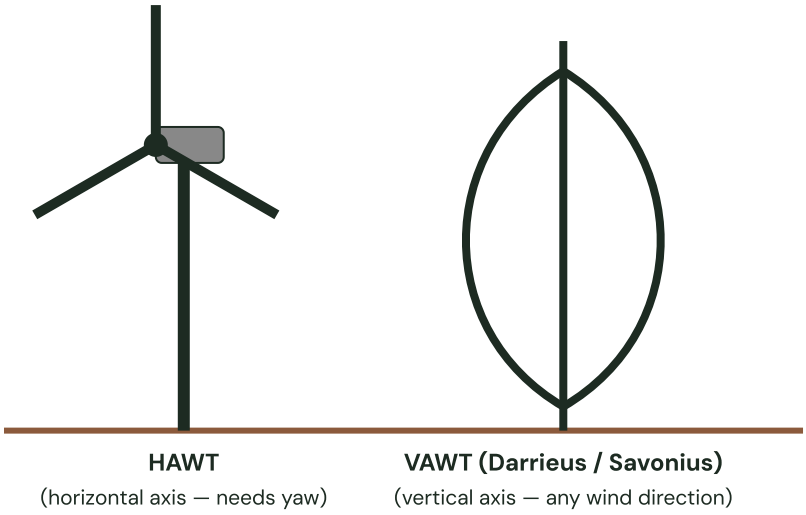
UNIT 7 · TOPIC 5

Types of Wind Mills and Their Applications

Syllabus topic: Types of wind mills and their applications

The wind-power equation and the Betz limit



Wind power in the air
P = ½ ρ A V³
Extractable: P = ½ ρ A V³ · Cp
BETZ LIMIT
Cp(max) = 16/27 = 0.593
Real machines: Cp ≈ 0.35–0.45

Wind machines and the wind-power equation (redrawn from the source figure)

Kinetic power in the wind: $P = \frac{1}{2} \cdot \rho \cdot A \cdot V^3$ ($\rho \approx 1.2$ kg/m^3; A = swept area)

Extractable power: $P = \frac{1}{2} \cdot \rho \cdot A \cdot V^3 \cdot Cp$ (Cp = power coefficient)

- **BETZ LIMIT:** maximum theoretical Cp = **16/27 = 0.593** — no turbine can extract more than ~59.3% of the wind's kinetic energy. Real machines reach Cp ~**0.35–0.45**.
- P ∝ V³ — DOUBLING the wind speed gives EIGHT times the power, so site wind speed is decisive. P also ∝ A ∝ (rotor diameter)².

Tip-speed ratio λ = blade-tip speed ÷ wind speed

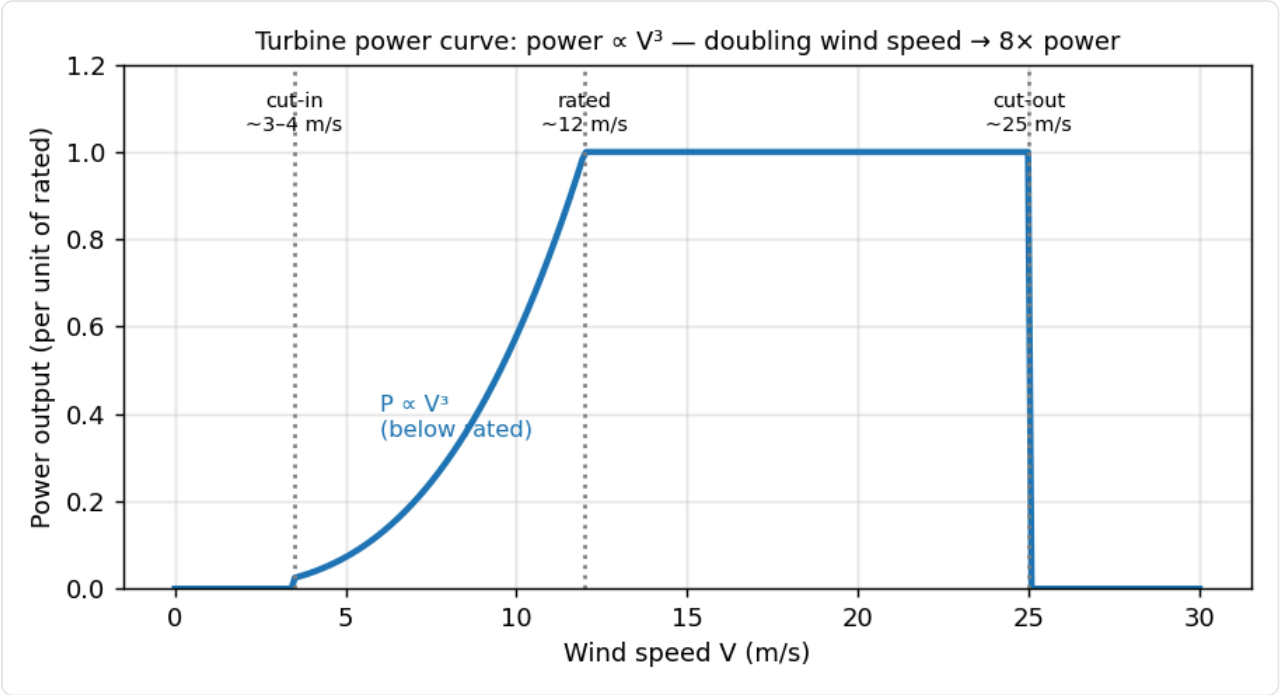
- High-speed, few-bladed rotors (2–3 blades) suit electricity generation.
- Low-speed, multi-bladed rotors give high starting torque for water pumping.

Types of wind machines

Type	Examples	Features
HORIZONTAL-AXIS (HAWT)	Propeller type (power generation); multi-blade "American" farm windmill (reciprocating water pumps)	Axis parallel to the ground/wind; high efficiency; needs YAW to face the wind
VERTICAL-AXIS (VAWT)	DARRIEUS ("eggbeater", lift-type, high speed); SAVONIUS (drag-type S-rotor, high torque, self-starting)	Accepts wind from ANY direction (no yaw); generator at ground level; generally less efficient

Classed by force: LIFT-type (propeller, Darrieus — efficient) vs DRAG-type (Savonius, cup — simpler).

Applications and operating speeds



Turbine power curve — power ∝ V³ below rated speed

- **Applications:** grid/standalone electricity (wind-electric generators, WEG), WATER PUMPING (irrigation/drinking), battery charging, wind–diesel and wind–solar hybrids.

Speed	Value	Meaning
CUT-IN	~3–4 m/s	Turbine starts
RATED	~11–14 m/s	Full output
CUT-OUT	~25 m/s	Shut down for protection

- Economic wind farming generally needs a mean annual wind speed above ~**5 m/s** at hub height.
- India's best wind states: Tamil Nadu, Gujarat, Karnataka, Maharashtra, Rajasthan, Andhra Pradesh.
- **Installed wind capacity ≈ 56.0 GW** on 31 March 2026 (57.44 GW by 30 June 2026) — India is 4th in the world.

Practice MCQs

Q71. The power available in wind of speed V through area A is:

(a) ½ρAV² (b) ½ρAV³
(c) ρAV (d) ρA²V

Q72. Wind power is proportional to wind speed raised to the power:

(a) 1 (b) 2
(c) 3 (d) 4

Q73. Doubling the wind speed increases available power by:

(a) 2× (b) 4×
(c) 8× (d) 16×

Q74. The Betz limit (maximum power coefficient) is:

(a) 0.35 (b) 0.50

(c) 0.593 (16/27) (d) 1.0

Q75. Wind power is also proportional to:

(a) Rotor diameter (b) Rotor diameter squared (swept area)

(c) Blade number (d) Tower height cubed

Q76. Tip-speed ratio is the ratio of:

(a) Wind speed to rotor rpm (b) Blade-tip speed to wind speed

(c) Power to torque (d) Diameter to length

Q77. The classic multi-blade 'American' farm windmill is mainly used for:

(a) Grid power (b) Water pumping (high starting torque)

(c) Battery charging only (d) Lighting

Q78. A HAWT (horizontal-axis turbine) requires which mechanism to face the wind?

(a) Yaw (b) Pitch only

(c) None (d) Brake

Q79. The Darrieus and Savonius rotors are:

(a) HAWTs (b) VAWTs (vertical axis)

(c) Water turbines (d) Steam turbines

Q80. A Savonius rotor is a:

(a) Lift-type high-speed rotor (b) Drag-type, high-torque, self-starting rotor

(c) HAWT (d) Gas turbine

Q81. An advantage of VAWTs is that they:

(a) Are most efficient (b) Accept wind from any direction (no yaw)

(c) Need no support (d) Have no generator

Q82. Typical cut-in wind speed of a turbine is about:

(a) 0.5 m/s (b) 3–4 m/s

(c) 12 m/s (d) 25 m/s

Q83. The cut-out speed at which a turbine shuts down for safety is about:

(a) 5 m/s (b) 12 m/s

(c) 25 m/s (d) 50 m/s

Q84. Real wind turbines achieve a power coefficient of about:

(a) 0.05–0.10 (b) 0.35–0.45

(c) 0.60–0.70 (d) 0.90

Q85. India's installed wind capacity (2026) is about:

(a) 20 GW (b) 40 GW

(c) 56 GW (d) 100 GW

Answers

Q71 — (b) ½ρAV³ — *$P = \frac{1}{2}\rho A V^3$.*

Q72 — (c) 3

Q73 — (c) 8× — *$P \propto V^3$: $2^3 = 8$.*

Q74 — (c) 0.593 (16/27) — *$C_p \le 16/27 = 0.593$ (Betz).*

Q75 — (b) Rotor diameter squared (swept area) — *$P \propto A \propto D^2$.*

Q76 — (b) Blade-tip speed to wind speed

Q77 — (b) Water pumping (high starting torque)

Q78 — (a) Yaw
Q79 — (b) VAWTs (vertical axis)
Q80 — (b) Drag-type, high-torque, self-starting rotor
Q81 — (b) Accept wind from any direction (no yaw)
Q82 — (b) 3–4 m/s
Q83 — (c) 25 m/s
Q84 — (b) 0.35–0.45
Q85 — (c) 56 GW — *India wind ≈ 56 GW (31 Mar 2026), 4th largest globally.*

Sources: Rai — Non-Conventional Energy Sources; Sukhatme; Duffie & Beckman; MNRE Bio-Energy & Wind programmes and PIB releases (as given in the source notes, July 2026); Betz (1919) actuator-disc theory; NPTEL renewable-energy & biomass lectures.